Bengü Süeda Şengül

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University of Illinois Urbana-Champaign

Research Interests

My research focuses on lipid-peptide hybrid membrane systems, biomimetic membranes, and the biophysical characterization of soft matter. I characterize membranes using atomic force microscopy (AFM), small-angle X-ray scattering (SAXS), confocal laser scanning microscopy (CLSM), and cryo-electron microscopy. My work aims to understand structure, mechanical properties, and functional properties of these systems, with broader implications for biomimetic cell membrane design and selective separation membranes.

Education

08/2023 – present
Ph.D. Materials Science and Engineering
University of Illinois Urbana-Champaign, IL, USA
GPA: 3.78 / 4.00
Advisor: Cecilia Leal

09/2019 – 06/2023
B.Sc. Materials Science and Nano Engineering
Sabanci University, Istanbul, Turkiye
Minor in Chemistry; Minor in Physics
GPA: 3.9 / 4.0
Dean's High Honour List; Full scholarship awarded for academic excellence
Advisor: Gozde Ozaydin Ince

09/2015 – 06/2019
IB Diploma Programme (38/45 points)
Ari Private High School, Ankara, Türkiye

Research Experience

08/2023 – present
Research Assistant, University of Illinois Urbana-Champaign, IL, USA
Advisor: Cecilia Leal
Researching thermally actuated lipid-peptide hybrid membranes using AFM, X-ray scattering, CLSM, and Cryo-EM. Investigating hybridization, mechanical properties, and ionic conductivity.

05/2026 – 08/2026
Visiting Research Assistant, Karolinska Institutet, Stockholm, Sweden
Advisor: Erdinc Sezgin
Evaluating the mechanism and sensitivity of smart probes for extracellular vesicles (EVs) and giant plasma membrane vesicles (GPMVs) imaging using fluorescence correlation spectroscopy (FCS) and single particle profiler (SPP).

09/2022 – 06/2023
Graduation Project, Sabanci University, Istanbul, Turkiye
Advisor: Gozde Ozaydin Ince
Designed and conducted a two-semester project on polymer nanoparticle synthesis for drug delivery and laser-induced photothermal therapy. Performed polymer nanoparticle synthesis, characterization, laser stimulation, and cell culture experiments.

Research Experience (continued)

09/2021 – 06/2023 **Lab Research Assistant,** Sabanci University Nanotechnology Research and Application Center, Istanbul, Turkiye
Advisors: Gozde Ozaydin Ince, Ozlem Kutlu
"Patch Synthesis for Breast Cancer Treatment" project (10 hrs/week). Performed polymer patch synthesis by blow spinning, iCVD coating, and cell culture experiments (live/dead cell assays).

06/2022 – 09/2022 **Research Intern (Erasmus+),** Institute of Materials Science of Barcelona (ICMAB-CSIC), Barcelona, Spain
Advisor: Anna Laromaine
Characterized bacterial cellulose composites by FTIR, SEM, fluorescence microscopy, and performed optical sensitivity experiments on composites. Gained experience in bacterial passages and silk production.

02/2022 – 06/2022 **PURE Undergraduate Researcher,** Sabanci University, Istanbul, Türkiye
Advisor: Gullu Kiziltas Sendur
Demonstrated 3D printing of molds and biosensor production by using conductive polymers.

06/2021 – 08/2021 **Research Intern (Erasmus+),** Italian Institute of Technology (IIT), Pisa, Italy
Advisor: Gianni Ciofani
Synthesized and characterized cell-membrane-coated polymer nanoparticles for the treatment of *glioblastoma multiforme* via SEM, DLS, and confocal microscopy. Learned cell culturing of human cells and polymer synthesis techniques.

03/2023 – 06/2023 **Design Engineer,** Turkish Aerospace Industries (TAI), Istanbul, Turkiye
Trained in structural design and characterization of aerospace vehicle parts.

Publications

[1] **B. S. Sengul**, N. Kambar, Y. Chen, Y. K. Go, J. Daye, C. M. Evans, and C. Leal, "Thermally actuated insertion of amphiphilic peptides into phospholipid bilayers," *ACS Nano (accepted, in revision)*, 2026.

[2] Y.-J. Chen, Y.-H. Tsao, **B. S. Sengul**, M.-H. Lin, C. Leal, C. M. Evans, and C. E. Sing, "Dynamic covalent crosslinking delays phase separation in networks of incompatible polymers," *Macromolecules*, vol. 59, no. 18, pp. 10 131–10 146, Sep. 2026, ISSN: 0024-9297. DOI: 10.1021/acs.macromol.6c00726.

[3] M. Porras-Gómez, **B. S. Sengul**, N. Kambar, S. Gluck, K. Flatt, C. Parry, C. Ruiz Le-Bert, D. Hernández-Saavedra, and C. Leal, "Interplay between pulmonary membrane properties and lung disease: A study of seven bottlenose dolphins," *eLife*, 2026. DOI: 10.7554/elife.109233.1.

[4] T. Kumarage, Y. Li, **B. S. Sengul**, M. B. Mustafa, J. Lou, M. D. Best, C. M. Schroeder, and C. Leal, "Ionizable lipids promote curvature remodeling and altered fluctuation dynamics in endosomal membranes," *bioRxiv*, 2026. DOI: 10.64898/2026.08.17.745287.

[5] E. Güney, J. Karimzadeh Khoei, **B. S. Sengul**, F. Can, and G. Ozaydin Ince, "Functional copolymer coatings on electrospun fibers via photo-initiated chemical vapor deposition," *Journal of Applied Polymer Science*, vol. 141, no. 15, e55210, 2024. DOI: https://doi.org/10.1002/app.55210.

[6] A. C. Egil, A. Carmignani, M. Battaglini, **B. S. Sengul**, E. Acar, G. Ciofani, and G. Ozaydin Ince, "Dual stimuli-responsive nanocarriers via a facile batch emulsion method for controlled release of rose bengal," *Journal of Drug Delivery Science and Technology*, vol. 74, p. 103 547, 2022, ISSN: 1773-2247. DOI: https://doi.org/10.1016/j.jddst.2022.103547.

Awards & Grants

2026
- **Registration Fee Waiver, Biophysical Society Conference**
- **Outstanding Laboratory Safety Recognition** – Featured in UIUC Division of Research Safety newsletter.

2024 – 2025
- **JUAMI Charz Proposal Competition Award ($1,000 research grant)** (USA–Tanzania collaboration)
  Collaborated with Raymond Sono Mahega (Tanzania) on the project titled "Investigation of the Influence of Biochar Characteristics on the Growth Performance of Spinach Under Wastewater Irrigation."
  Featured in the Materials Science Department newsletter, including an invited interview.

2025
- **List of Excellent Teachers – Outstanding TA**, Materials Science Department, UIUC
- **Travel Grant ($500), International Conference of Neutron Scattering**, Awarded for participation; unable to attend.

2023
- **Full Scholarship for Academic Excellence**, Sabanci University

2019-23
- **Dean's High Honour List**, Sabanci University

2022
- **Erasmus+ Internship Grant**, Barcelona (ICMAB-CSIC)

2021
- **Erasmus+ Internship Grant**, Pisa (IIT)

Workshops

09 – 11/2025
- **Designing & Delivering STEM Outreach Workshop**, UIUC.
  Designed and led STEM activity for K–12 students on National STEM Day.

05/2025
- **Mentoring Matters Workshop**, UIUC.
  Earned UIUC Graduate College Mentoring Certificate.

01/2025
- **AFM Bootcamp & Symposium**, Materials Research Laboratory, UIUC.
  Presented the AFM results from my research.
  Hands-on training in AFM imaging, optimization, data analysis, and liquid-phase AFM.

10/2024
- **Bio-SANS Workshop**, Oak Ridge National Lab (online).
  Learned biological SANS data analysis and presented my research.

05/2024
- **Advanced Materials Characterization Workshop (AMC2024)**, Materials Research Laboratory, UIUC.

04/2022
- **Circular Engineering Across Disciplines (CEAD) Training Week**, KU Leuven–UHasselt, Belgium.
  Planned workshops and presented outcomes on circular economy and sustainable materials.

Conferences & Talks

ACS POLY/PMSE'26
- **Platform Talk:** "Thermally Actuated Insertion of Amphiphilic Peptides into Phospholipid Bilayers", American Chemical Society: the Division of Polymer Chemistry (POLY) and the Division of Polymeric Materials: Science and Engineering (PMSE) Summer Symposium at UIUC, Urbana, IL.

BPS'26
- **Platform Talk and Poster:** "Thermal Actuation Drives Hexagonal Peptide Domain Formation in Peptide-Polymer Amphiphile and Lipid Hybrid Membranes", Biophysical Society Annual Meeting, San Francisco, CA.

BPS'25
- **Poster:** "Temperature Responsive Out-of-Plane Correlations of Phase Separated Domains in Lipid Membranes Hybridized with Polypeptide Copolymers", Biophysical Society Annual Meeting, Los Angeles, CA.

Conferences & Talks (continued)

- MRS'24 — Attendee, Materials Research Society Spring Meeting, Seattle, WA.
- E-MRS'22 — Presenter: Guney, E., Ustunkaya, B., **Sengul, B. S.**, Ozaydin Ince, G., & Kutlu O. "Coating of functional patches for use in biomedical applications." European Materials Research Society Virtual Conference.
- 2022 — Attendee, Workshop on Simulating Soft Matter with Dissipative Particle Dynamics, Istanbul, Türkiye.
- 2019 — Attendee, METU 3rd Astrobiology Conference, Ankara, Türkiye.
- 2018 — Attendee, METU 2nd Astrobiology Conference, Ankara, Türkiye.

Service & Outreach

- 08/2026 – present — **Graduate Student Ambassador**
  Peer mentor for the incoming Grainger engineering graduate students at UIUC.
  Helping the College of Engineering with student orientations and recruitment.
- 09/2024 – present — **Designated Laboratory Safety Contact**, UIUC.
  Led safety initiatives recognized as "out-of-reports" for the first time in decades in the department; featured in UIUC Division of Research Safety newsletter.
- 03/2025, 03/2026 — **Panelist** Incoming MatSE PhD Student Orientation Panel, UIUC.
- 10/2024 – 12/2025 — **Graduate Student Representative**, MatSE Community Committee, UIUC.
  Held weekly coffee&chat sessions with colloquium speakers from different universities.
  Planned and organized a MatSE community town hall.
- 08/2024 – 08/2025 — **President, Turkish Student Association (TSA)**, UIUC.
  Raised $3,500 for national day celebrations, fundraisers, and cultural events.
- 09/2021 – 06/2023 — **Erasmus Buddy**, Sabanci University.
  Mentored exchange students adapting to life in Turkiye.
- 09/2022 – 12/2023 — **Co-organizer, "Teachers are Human" Radio Program**, Sabanci University Radio Club.
  Hosted weekly episodes featuring faculty members.

- 02/2022 — **Erasmus+ Project: Script Action for the Future Earth**, Hungary.
  Attended a project about environmental pollution
  Surveyed and interviewed around 50 people about pollution/climate change, presented the findings
- 02/2020 — **Erasmus+ Project: Walls Of Ignorance**, Slovakia.
  Participated in a project about human rights and debated Turkiye's situation on an international platform.

Teaching & Mentoring

- 09/2024 – present — Supervising BS student Shane Spatol (UIUC) on biomimetic membrane synthesis and characterization.
- Spring 2025 — Teaching Assistant, Biomolecular Materials Science (MSE 473), UIUC.
  Selected for the *List of Excellent Teachers* as Outstanding TA in the Materials Science Department.
- Summer 2024 — Supervised junior student Joanie Daye (UIUC, now a Master's student in Bioengineering, UIUC) on lipid-peptide hybridization and characterization.

Teaching & Mentoring (continued)

07/2022	Supervised high school student Blai Higueras (ICMAB, Spain) on bacterial cell culture, silk purification, and bacterial composite synthesis.
Spring 2022	Supervised four freshman students (one is a Master's student at UCL) at Sabanci University on nanoparticle synthesis and characterization.
09/2020 – 06/2021	Teaching Assistant, Calculus I & II, Sabanci University

Skills

Languages	Turkish (Native), English (Advanced, TOEFL iBT: 100), Spanish (Intermediate, DELE B1)
Laboratory	Atomic force microscopy, Small/wide-angle X-ray scattering, Cryogenic electron microscopy, fluorescence correlation spectroscopy, confocal microscopy, single particle profiler, small angle neutron scattering, blow spinning, electrospinning, iCVD, cell culture
Programming	Python, MATLAB
Data Analysis	AutoCAD, Gwyddion, Igor, BioXTAS RAW